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# Summative Course Assessment

**Assignment details**

**Due** Jan 19, 11:59 PM IST

**Attempts** 0 left (1 attempts every 8 hours)

Retry

Submitted Jan 14, 10:11 PM IST

Your grade 93.33%

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**1.** What is the main benefit of using transfer learning in deep learning network architectures? **1 / 1 point**

- It allows for deeper networks with no impact on computation.
- It dynamically adjusts the learning rate of layers during training for efficiency.
- It increases the network's complexity and computational requirements.
- It decreases the need for pre-training a model.

Correct
**Correct!**

**2.** What challenge is highlighted by the statement "Deep Neural Networks are Easily Fooled"? **1 / 1 point**

- DNNs can overfit to the training data.
- DNNs require large amounts of data to learn.
- DNNs can misclassify images that are slightly altered or contain noise.
- DNNs are computationally efficient.

Correct
Correct!

**3.** What does the word hash technique address in the context of word representation for neural networks? **1 / 1 point**

- It reduces the dimensionality of word vectors.
- It increases the accuracy of neural networks.
- It encrypts word vectors for security.
- It enhances the color representation of images.

Correct
**Correct.**

**4.** Which of the following applications can benefit from the transformer model? **1 / 1 point**

- Image recognition only
- Sequence-to-sequence tasks like machine translation
- Simple classification tasks
- Processing static data structures

Correct
**Correct!**

**5.** In image transfer learning, which layers are typically copied from the source model? **1 / 1 point**

- The last few layers
- The first few layers
- Only the middle layers
- No layers are copied; they are all retrained

Correct
**Correct!**

**6.** What is the primary benefit of weight clustering in network compression? **1 / 1 point**

- To improve the training time of the network.
- To cluster similar weights, thereby reducing the model size.
- To increase the number of parameters for better accuracy.
- To expand the size of the network.

Correct
**Correct!**

**7.** Why is conservative training important in Transfer Learning? **1 / 1 point**

- It prevents the model from learning anything new
- It ensures rapid training on new tasks
- It helps in avoiding overfitting when adapting to new, limited data
- It allows for the use of unlabeled data only

Correct
**Correct!**

**8.** Why are CNNs efficient for image recognition tasks? **1 / 1 point**

- They require the entire image to detect patterns.
- Each neuron is connected to every pixel in the image.
- Patterns can be detected using small regions of the image, requiring fewer parameters.
- CNNs use the same parameters for different layers.

Correct
Correct!

**9.** What is a key advantage of using self-attention over recurrent neural networks (RNNs) for processing sequences? **1 / 1 point**

- Self-attention requires less training data.
- Self-attention can capture dependencies regardless of distance in the sequence more effectively.
- Self-attention uses fewer parameters. (Can be true in some cases, depends on architecture)
- Self-attention operates on fixed-length sequences.

Correct
**Correct!**

**10.** What does 'embedding space' in Zero-shot Learning represent? **1 / 1 point**

- A physical space where data is stored
- A conceptual space where similar attributes are closer together
- The total memory used by the neural network
- A database of all possible attributes

Correct
**Correct!**

**11.** What role does the softmax function play in a neural network? **1 / 1 point**

- It acts as a regularization technique.
- It speeds up matrix operations.
- It serves as a multi-class classifier at the output layer.
- It initializes the weights of the neural network.

Correct
**Correct.**

**12.** The Lottery Ticket Hypothesis in network compression suggests that: **0 / 1 point**

- Every network has a winning combination of weights from the start
- Pruned networks are always superior to unpruned networks
- Initial random weights are not significant for network training
- Keeping the sign of the initial weights is critical after pruning

Incorrect
Incorrect. The hypothesis suggests there might be a good sub-network within a randomly initialized network, not that all networks have one.

**13.** What determines the number of channels in the output after a convolution operation in a CNN? **1 / 1 point**

- The size of the filters.
- The number of filters applied.
- The stride with which the filters are applied.
- The padding added to the input image.

Correct
Correct!

**14.** What is one of the suggested solutions when achieving good results on training data but not on testing data? **1 / 1 point**

- Increase the network size
- Remove all regularization methods
- Implement early stopping, regularization, and dropout
- Always use a larger learning rate

Correct
Correct!

**15.** Which of the following best describes the transformer model's scalability compared to traditional sequence processing models? **1 / 1 point**

- Transformers can only scale horizontally.
- Transformers are less scalable due to higher computational requirements. (Can be true for very large models)
- Transformers offer better scalability due to parallelization of self-attention.
- Transformers scale similarly to RNNs but with higher accuracy.

Correct
**Correct!**